

Building a Centralized Compliance Platform for Global Trade Surveillance

Introduction

A US-based global asset management firm managing over \$500 billion in assets partnered with NuWare to modernize its Legal and Compliance infrastructure. The client's compliance team was responsible for monitoring global trading activities, ensuring regulatory adherence, and managing investigations into historical trading orders and executions.

With trade data dispersed across multiple systems and geographies, the compliance team faced significant challenges in data reconciliation, accuracy, and timely reporting. NuWare designed and implemented a centralized Compliance Data Warehouse System, enabling unified access to trade, order, and market data across all regions; empowering the compliance team to deliver faster, more accurate, and auditable regulatory reporting.

The Challenge

The firm's global footprint and trading complexity created significant obstacles for compliance operations.

Key Challenges:

1. **Fragmented Data Sources:**

Trade and order data originated from multiple order management systems (OMS) and execution venues across different countries, each with varying formats and time zones.

2. **High Data Volume:**

The compliance team needed to process and analyze approximately 500 million transactions per day, including order, route, and execution data, along with market quotes from various vendors.

3. **Complex Reconciliation Requirements:**

Compliance analysts were required to match order data against market data to assess best execution quality and generate regulatory reports for multiple authorities.

4. **Manual Processes and Delays:**

Manual reconciliation led to slower turnaround times, inconsistencies, and

operational inefficiencies in responding to internal and external information requests.

5. Scattered Historical Data:

Limited access to historical records hindered long-term trade pattern analysis and regulatory investigations.

NuWare's Approach

NuWare engineered a comprehensive Compliance System by implementing an Enterprise Data Warehouse (EDW) architecture. The solution seamlessly aggregated, standardized, and analyzed trade data from multiple global sources, enabling high-speed regulatory reporting and data-driven decision-making.

1. Centralized Data Repository

- Designed a scalable data warehouse integrating trade, order, and market data from global OMS systems and vendors.
- Unified disparate data formats and synchronized time zones across geographies to ensure consistency and accuracy.
- Supported T+1 data availability and maintained access to 10+ years of historical data for audits and investigations.

2. ETL and Data Standardization

- Developed a custom ETL process using SSIS to load, transform, and validate massive data volumes daily.
- Automated reconciliation between trade orders and market quotes, reducing manual intervention.
- Implemented data compression techniques to reduce storage and maintenance costs while ensuring fast retrieval.

3. Reporting and Visualization Layer

- Built interactive dashboards and ad-hoc reporting tools using SSRS for real-time data querying and compliance analytics.
- Introduced an Advanced Query Search feature, enabling investigators to retrieve any transaction record instantly.
- Provided internal audit and compliance teams with pre-built and customizable regulatory report templates.

4. Application Development and User Experience

- Developed a .NET Windows application for secure, user-friendly data access and visualization.
- Integrated hierarchies and tree structures in dimensions to allow users to traverse entire order flows intuitively.

5. Security and Data Governance

- Established robust data access controls and audit trails to ensure compliance with global regulatory standards.
- Integrated historical order and trade information with market data for enhanced transparency and traceability.

Technologies Used

Component	Technology Stack
Database	Azure SQL Server 2019, Sybase IQ
ETL Framework	SQL Server Integration Services (SSIS)
Reporting Tools	SQL Server Reporting Services (SSRS)
Application	Microsoft .NET (Windows Platform)

Outcomes

The implementation of NuWare's Compliance System delivered significant operational efficiency, transparency, and accuracy for the client's global compliance operations.

1. Centralized Data and Unified Access

- Consolidated trade, order, and market data from multiple systems into one centralized platform.
- Enabled cross-functional teams such as Middle Office and Risk to leverage the same data for risk assessment and P&L calculations.

2. Increased Efficiency and Speed

- Automated reconciliation processes improved turnaround time for regulatory reporting.
- Reduced manual intervention and human error across compliance operations.

3. Enhanced Data Accuracy and Timeliness

- Delivered T+1 availability of accurate, reconciled data for all transactions.
- Improved responsiveness to ad-hoc requests and regulatory inquiries.

4. Improved Audit and Investigative Capabilities

- Provided access to 10+ years of historical data, facilitating long-term compliance investigations.
- Enabled comprehensive audit trails for all order and execution events.

5. Cost-Effective and Scalable Infrastructure

- Compressed data storage reduced maintenance costs.
- The scalable architecture supported continuous growth in data volume without performance degradation.

Future Outlook

Following the successful deployment of the compliance system, NuWare continues to collaborate with the client to:

- Integrate AI-powered anomaly detection for identifying trading irregularities.
- Expand the platform to include real-time regulatory alerting and surveillance analytics.
- Explore cloud-native scalability for global accessibility and performance optimization.

These initiatives aim to evolve the system into a next-generation intelligent compliance platform, ensuring proactive risk management and complete transparency.